Sampling Distributions and Analysis of Variance Report

The ZZZ Factor: A Report on Factors Influencing Sleep Quality in Students

A report submitted to SVKM'S Narsee Monjee Institute of Management Studies as part of Semester III of B.Sc. Applied Mathematical Computing

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1 Introduction

Sleep is a fundamental biological process, crucial for cognitive function, memory consolidation, and overall physical and mental well-being. For students navigating demanding academic and social environments, consistent, high-quality sleep is not a luxury, but a necessity for achieving success and maintaining well-being. It serves as the foundation for learning, focus, and emotional resilience, directly influencing their ability to perform, adapt, and thrive in their academic pursuits.
In this report we employ statistical hypothesis testing methodology to assess sleep quality and factors influencing sleep quality. In doing so we aim to provide data driven insights on sleep patterns in students which can help to foster healthy practices.

2 About the Dataset

The dataset used in this analysis comprises a comprehensive collection of data collected from students on various metrics on sleep quality as well as on various lifestyle and academic factors that are linked to sleep quality.

2.1 Source

The data has been collected as primary data directly from students of SVKM institutions by way of a questionnaire circulated via Google Forms. The questionnaire has been attached below.

2.2 Questionnaire

Table 1: Questionnaire on Sleep Quality and Lifestyle

Variable	**Question**
Age	Age
Gender	Gender
Program/Course	Program/Course
Sleep Duration	On average, how many hours of sleep do you get per night?
Sleep Latency	How long does it typically take you to fall asleep?
Night Awakenings	On an average night, how many times do you wake up?
Bedtime Weekday	What is the average time you go to bed on a weekday (24-hour clock, e.g., 22:30)?
Wake Time Weekday	What is the average time you wake up on a weekday (24-hour clock, e.g., 06:00)?
Sleep Quality	On a scale of 1 to 10, with 1 being 'very poor' and 10 being 'excellent,' how would you rate your overall sleep quality?
Sleep Improvement	Are you taking a step for improving your sleep schedule?
Physical Activity Days	In a typical week, on how many days do you engage in moderate-to-vigorous physical activity (e.g., brisk walking, running, cycling, gym workouts, sports) for at least 30 minutes?

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Table 1 – continued from previous page

Variable	**Question**
Physical Activity Duration	On the days you are physically active, what is the average duration of physical activity?
Caffeine Consumption	How many caffeinated beverages (e.g., coffee, tea, energy drinks) do you consume per day, on average?
Caffeine Reason	What is the primary reason for your caffeine consumption?
Alcohol Consumption	Do you consume alcoholic beverages?
Alcohol Timing	On days when you consume these, approximately how close to your bedtime do you typically do so? (no of hours before the bedtime)
Commute Time	On a typical day, what is your total two-way travel time to and from your institute/college?
Academic Screen Time	On a typical day, approximately how many hours and minutes do you spend looking at screens for ACADEMIC purposes? (e.g., online classes, assignments, research)
Non-Academic Screen Time	On a typical day, approximately how many hours and minutes do you spend looking at screens (phone, laptop, tablet, TV) for NON-ACADEMIC purposes? (e.g., social media, entertainment, gaming)
Residence	Where do you currently reside?
Smoking/Tobacco	Do you currently smoke or use tobacco products (e.g., cigarettes, vapes, chewing tobacco)?
Health Problems	Do you have any health problems?

3 Summary

Table 2: Summary Statistics of Sleep Quality Survey Data (n=75)

Statistic	Age	Sleep Hours	Time to Sleep (h)	Wake-ups /night	Bed-time (24h)	Wake-up (24h)	Sleep Quality (1-10)	Active Days /week	Act. Dur. (h)	Caff. Drinks /day	Com-mute (h)	Acad. Screen (h)	Leisure Screen (h)
count	75.00	75.00	75.00	75.00	75.00	75.00	75.00	75.00	75.00	75.00	75.00	75.00	75.00
mean	18.77	6.53	0.79	1.19	22.22	19.65	5.71	3.69	1.66	1.48	1.46	3.16	5.01
std	1.69	1.70	1.81	1.15	5.37	10.37	2.44	2.37	1.75	1.39	1.38	2.72	3.36
min	16.00	3.00	0.00	0.00	10.00	7.00	1.00	0.00	0.00	0.00	0.00	0.00	1.00
25%	18.00	5.50	0.17	0.00	22.50	9.00	4.00	2.00	0.50	0.50	0.50	1.00	3.00
50%	18.00	6.50	0.42	1.00	24.00	26.00	8.00	3.00	1.00	1.00	1.00	3.00	4.00
75%	18.00	7.00	0.70	2.00	25.38	30.00	7.00	6.00	2.00	2.00	2.00	4.00	7.00
max	25.00	12.00	15.00	5.00	30.50	30.75	10.00	7.00	12.00	7.00	7.00	18.00	18.00

4 Graphical Representations

Sleep Latency

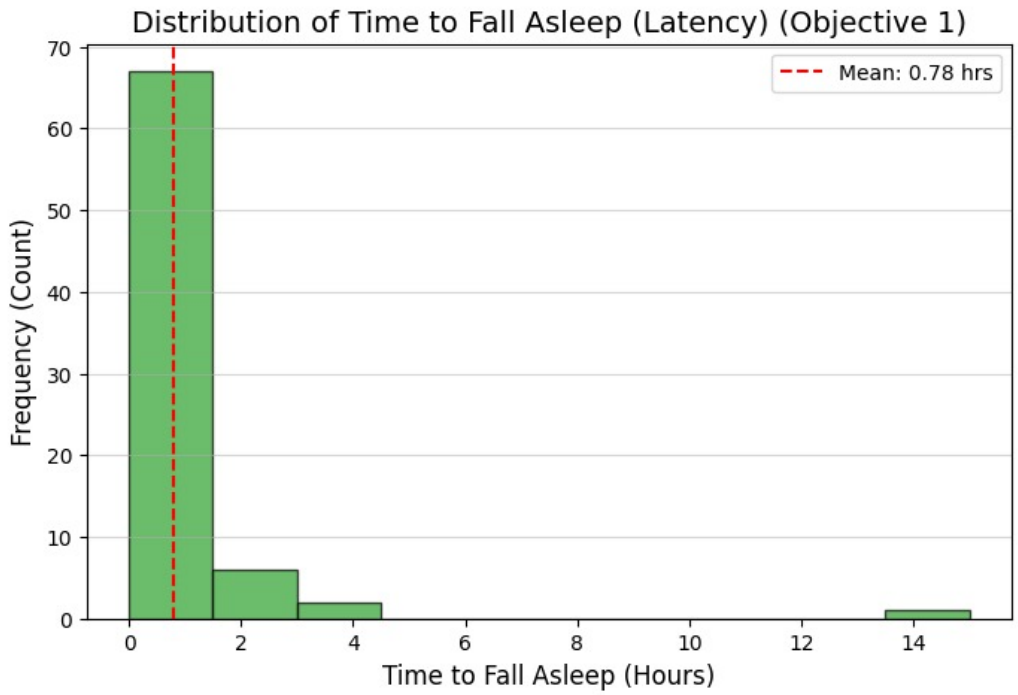

Figure 1: Distribution of Time to Fall Asleep (Sleep Latency) among Survey Participants

The histogram shows that the majority of students experience relatively quick sleep onset, with a mean latency of 0.78 hours (approximately 47 minutes). The distribution exhibits significant right-skewness, indicating that while most participants fall asleep within a reasonable time frame, a notable minority experiences prolonged sleep latency, potentially indicating sleep onset difficulties or insomnia symptoms in this subgroup.

Self Reported Sleep Quality

Figure 2: Distribution of Self-Reported Sleep Quality Ratings among Survey Participants

The histogram reveals a bimodal distribution with peaks at the lower and higher ends of the sleep quality scale.This polarization suggests two distinct subgroups within the student population: one experiencing poor sleep quality (ratings 2-4) and another reporting good to excellent sleep quality (ratings 8-10). The relatively low frequency in the middle range (5-7)indicates that students tend to perceive their sleep quality as either clearly problematic or clearly satisfactory, with few neutral assessments.

Relationship between Sleep Duration and Sleep Quality

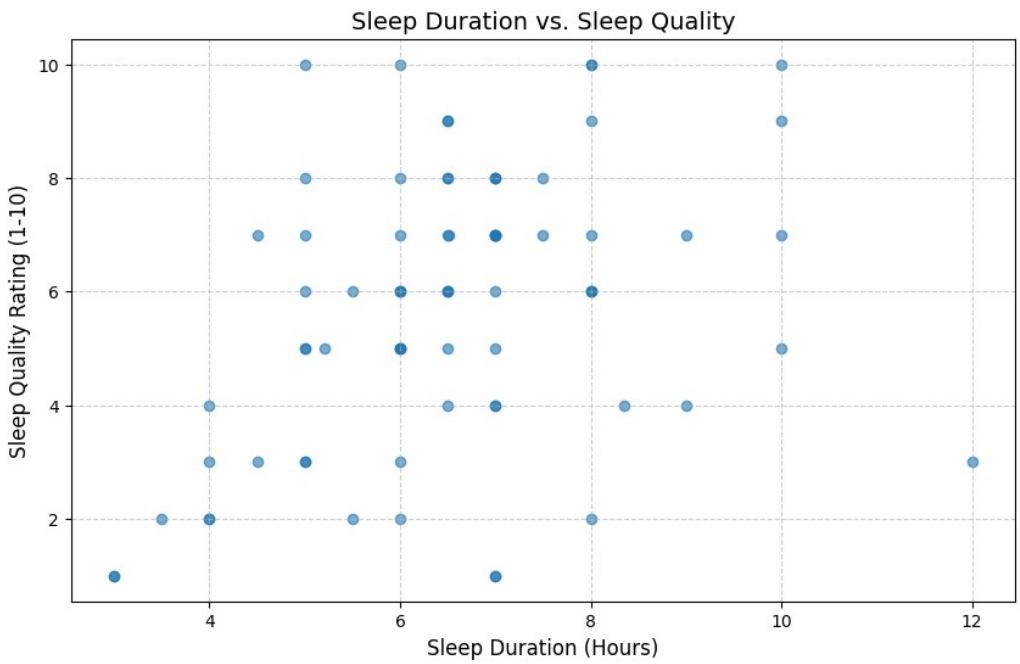

Figure 3: Relationship between Sleep Duration and Sleep Quality Ratings

This visualization examines how the quantity of sleep relates to perceived sleep quality. The scatter plot reveals a moderate positive correlation, suggesting that students who obtain longer sleep durations tend to rate their sleep quality higher. However, the relationship is not perfectly linear, indicating that sleep quality involves more than just duration. Notably, some students with adequate sleep duration (7-9 hours) still report lower quality scores, while others with shorter sleep durations report high satisfaction, highlighting the subjective nature of sleep quality assessment and the influence of additional factors such as sleep continuity, depth, and daytime functioning.

Relationship between Physical Activity and Sleep Duration

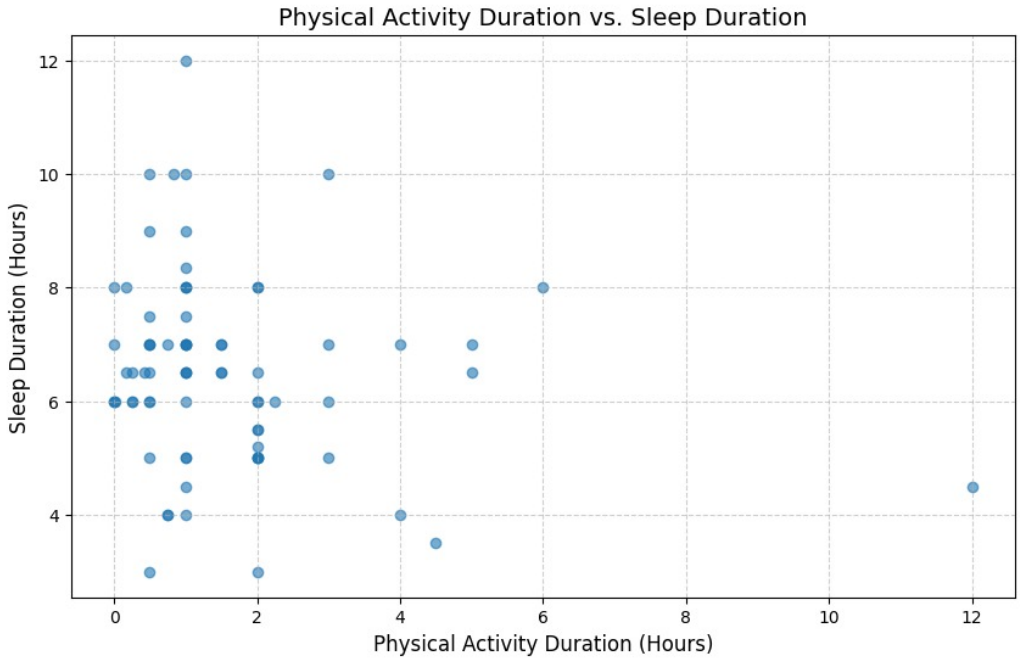

Figure 4: Relationship between Physical Activity Duration and Sleep Duration

The scatter plot examines the correlation between daily physical activity and sleep duration. Initial observation suggests a weak positive relationship, indicating that students who engage in longer physical activity sessions may tend to have slightly longer sleep durations. However, the scattered distribution also highlights significant individual variability, suggesting that physical activity duration alone does not strongly predict sleep duration, and other factors likely play important roles in determining sleep patterns.

Relationship between Caffeine Consumption and Sleep Duration

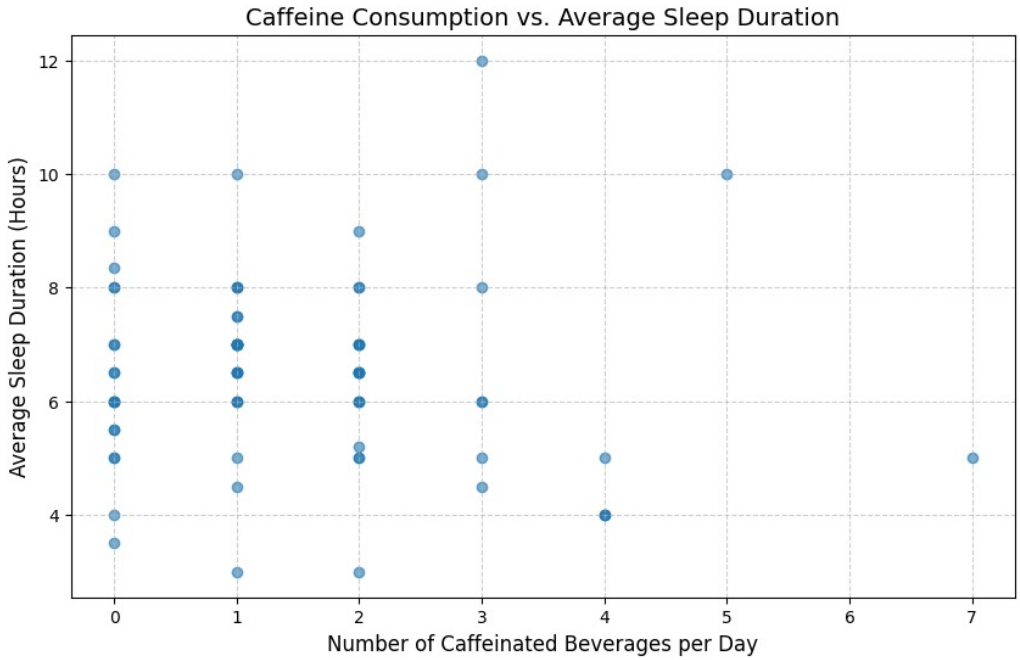

Figure 5: Impact of Caffeine Consumption on Average Sleep Duration

This visualization explores how daily caffeine intake relates to sleep duration. The data appears to show a weak negative correlation, where higher consumption of caffeinated beverages is associated with reduced sleep duration. This loosely aligns with established physiological knowledge that caffeine, as a stimulant, can interfere with sleep onset and reduce total sleep time. The trend suggests that students consuming 3 or more caffeinated drinks daily may be at risk for shorter sleep durations, potentially affecting academic performance and overall well-being.

Relationship between Screen Time and Sleep Duration

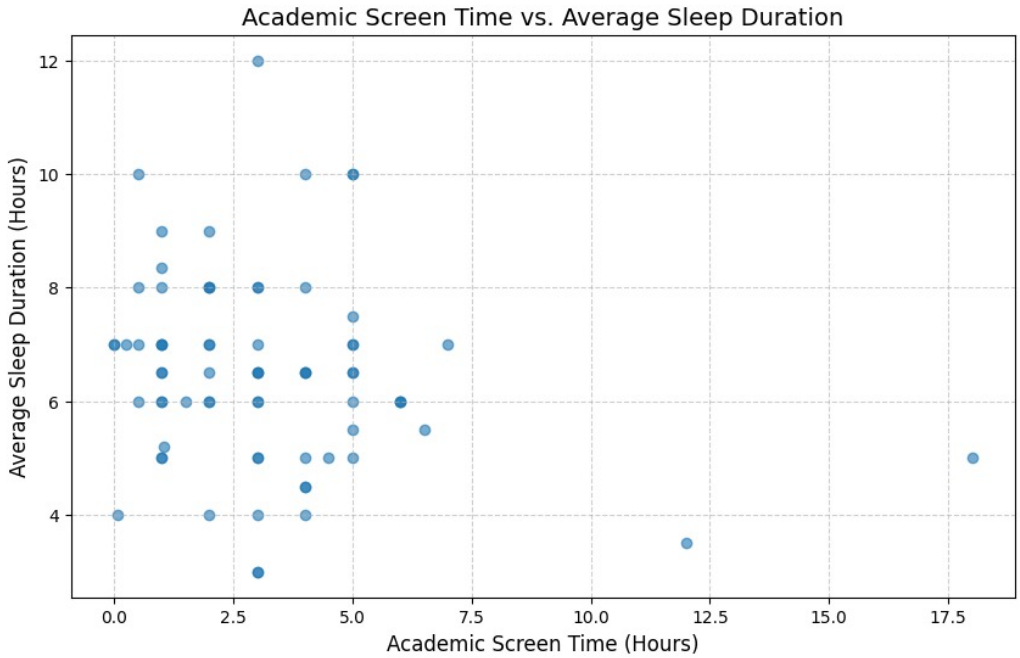

Figure 6: Relationship between Academic Screen Time and Average Sleep Duration

The scatter plot examines how time spent on academic screen activities correlates with sleep duration. The data shows a weak negative relationship, suggesting that excessive academic screen time may be associated with slightly reduced sleep duration. This could be attributed to several factors: late-night studying, academic stress, or the impact of blue light from screens on melatonin production and sleep onset. However, the wide dispersion of data points indicates that academic screen time alone is not a strong predictor of sleep duration, and individual time management strategies likely play a significant role in mediating this relationship.

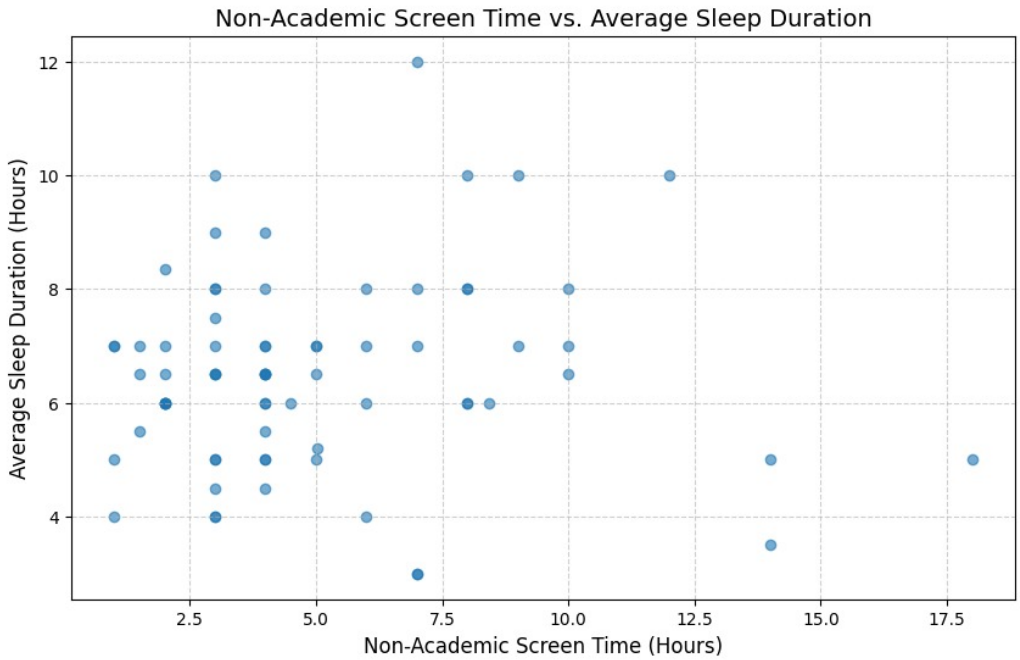

Figure 7: Relationship between Non-Academic Screen Time and Average Sleep Duration

The scatter plot examines the correlation between leisure screen time (social media, entertainment, gaming) and sleep duration. The data reveals a weak negative relationship, suggesting that increased non-academic screen time may be associated with slightly reduced sleep duration. This pattern could be attributed to several factors: late-night screen use displacing sleep time, the stimulating effects of entertainment content, or blue light exposure from devices affecting melatonin production and delaying sleep onset. However, the considerable scatter in the data indicates that non-academic screen time alone is not a strong predictor of sleep duration, and individual differences in screen usage habits and sleep resilience play important roles.

Relationship between Health Problems and Sleep Duration

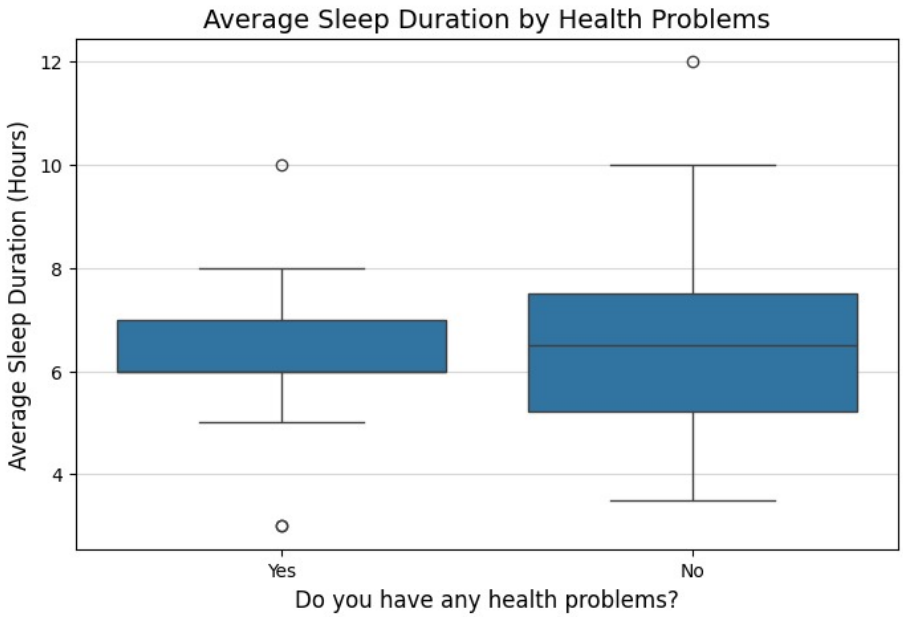

Figure 8: Comparison of Average Sleep Duration Between Students With and Without Health Problems

The bar chart reveals a notable difference in sleep duration based on health status. Students reporting health problems demonstrate significantly shorter average sleep duration compared to their healthy counterparts. This finding aligns with clinical observations that various health conditions can disrupt sleep patterns through pain, discomfort, medication side effects, or condition-specific symptoms. The substantial gap between the two groups underscores the importance of considering underlying health issues when addressing sleep complaints in the student population, and suggests that students with health problems may require targeted sleep interventions.

Relationship between Perceptions of Sleep Quality and Health Problems

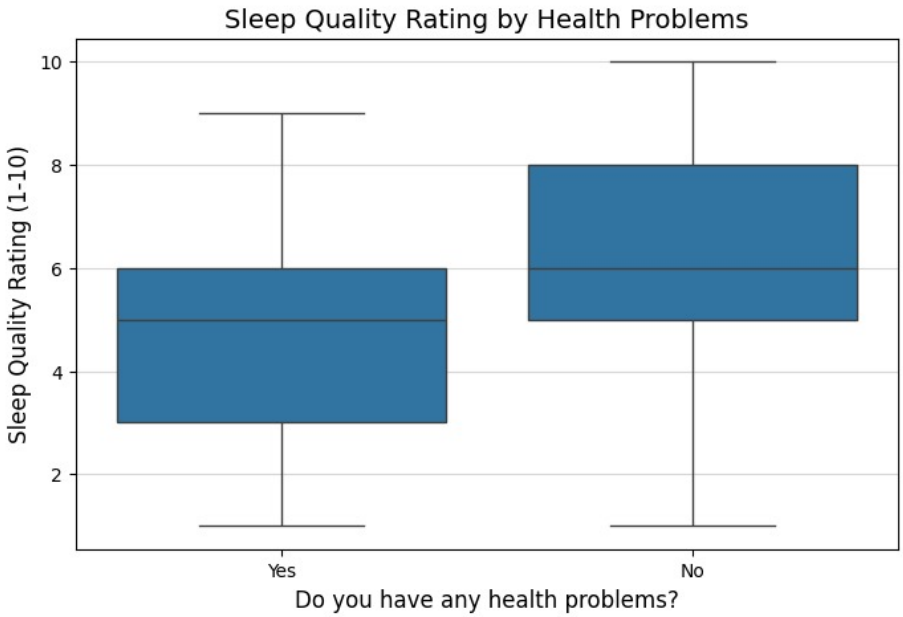

Figure 9: Comparison of Sleep Quality Ratings Between Students With and Without Health Problems

This visualization demonstrates a clear disparity in subjective sleep quality based on health status. Students with health problems report significantly lower sleep quality ratings compared to healthy students. The magnitude of this difference is substantial, indicating that health issues may profoundly impact how students perceive their sleep quality. This could result from various factors including pain disrupting sleep continuity, medication effects, psychological distress associated with health conditions, or direct physiological impacts of specific illnesses on sleep architecture. The finding emphasizes the bidirectional relationship between physical health and sleep quality in the student population.

5 Hypothesis Testing

5.1 Test 1: Comparison with Recommended Sleep Duration

- **Objective:** Determining whether average sleep differs significantly from recommended 8 hours
- **Hypothesis Test:** Single Mean t-test
- **Hypothesis:** H_0: Sample average sleep = 8 hours, H_1: Sample average sleep $\neq$ 8 hours
- **Results:** t = -7.5727, p-value < 0.001
- **Conclusion:** Reject H_0 - Average sleep is significantly less than recommended 8 hours

5.2 Test 2: Gender and Sleep Duration

- **Objective:** Determine if average sleep differs between male and female students
- **Statistical Test:** Independent t-test
- **Hypothesis:**
  - H_0: No significant difference in sleep duration between genders
  - H_1: Significant difference exists
- **Results:** $t = -1.3504$, p-value $= 0.1845$
- **Conclusion:** Fail to reject H_0 - No statistically significant gender difference in sleep duration

5.3 Test 3: Sleep Duration Across Academic Programs

- **Objective:** Determine if average sleep differs across major academic programs
- **Statistical Test:** ANOVA
- **Hypothesis:**
  - H_0: Average sleep does not differ across academic programs
  - H_1: Significant differences exist between programs
- **Results:** $F = 0.0023$, p-value $= 0.9977$
- **Conclusion:** Fail to reject H_0 - Sleep duration does not significantly differ by academic program

5.4 Test 4: Alcohol Consumption and Sleep Duration

- **Objective:** Determining whether average sleep differs with respect to alcohol consumption
- **Hypothesis Test:** Independent t-test
- **Hypothesis:** $H_0 : \mu_{\text{consumers}} = \mu_{\text{non-consumers}}$, $H_1 : \mu_{\text{consumers}} \neq \mu_{\text{non-consumers}}$
- **Results:** t = 0.6974, p-value = 0.4878
- **Conclusion:** Fail to reject H_0 - No significant difference in sleep duration between alcohol consumers and non-consumers

5.5 Test 5: Relationship between Caffeine Consumption and Sleep Duration

- **Objective:** To determine the nature of the relationship between sleep duration and caffeinated beverage consumption
- **Hypothesis Test:** Pearson's correlation
- **Hypothesis:** $H_0 : \rho = 0$, $H_1 : \rho \neq 0$
- **Results:** ρ = -0.06, p-value = 0.6120
- **Conclusion:** Fail to reject H_0 - No significant correlation between caffeine consumption and sleep duration

5.6 Findings and Conclusions

Test	**Statistical Test**	**Null Hypothesis (H_0)**	**Statistic**	**p-value**	**Conclusion**
Comparison with Recommended Sleep Duration	Single Mean t-test	Sample avg sleep = 8 hrs	$t = -7.57$	< 0.001	Reject H_0. Average sleep is significantly less than recommended 8 hours.
Gender and Sleep Duration	Independent t-test	No significant difference in sleep duration between genders	$t = -1.35$	0.1845	Fail to reject H_0. No statistically significant gender difference in sleep duration.
Sleep Duration Across Academic Programs	ANOVA	Average sleep does not differ across academic programs	$F = 0.0023$	0.9977	Fail to reject H_0. Sleep duration does not significantly differ by academic program.
Alcohol Consumption and Sleep Duration	Independent t-test	No significant difference between alcohol consumers and non-consumers	$t = 0.70$	0.4878	Fail to reject H_0. No significant difference in sleep duration between alcohol consumers and non-consumers.
Caffeine Consumption and Sleep Duration	Pearson's correlation	$\rho = 0$ (no correlation)	$\rho = -0.06$	0.6120	Fail to reject H_0. No significant correlation between caffeine consumption and sleep duration.

Table 3: Summary of Hypothesis Tests for Sleep Duration Study

- The data suggests that most students are unable to get the recommended 8 hours of sleep. This can be detrimental to their mental health and attention span.
- Surprisingly, the data suggests no significant relationship between alcohol and caffeine consumption and sleep duration, contrary to findings of medical research. This can be attributed to incorrect reporting by students and sample bias. It can also be due to other variables playing a larger role in affecting sleep patterns as compared to substance consumption.
- Screen time is weakly associated with reduced sleep duration, which is in conformity with

medical research on the topic.

- Self reported data on sleep quality suggests that most students are getting poor to medium quality sleep. Reduced sleep duration and academic stress are likely causes of this. Non academic sources such as social media and gaming screen time are also contributors to this.
- Most of the sample experiences sleep latency in the normal range, which is in accordance to medical research as high sleep latency is usually associated with aging and insomnia and unlikely to be caused by lifestyle factors.